PCBC H Parametric & Sequences Test (Practice B) **Name**_____ _____/16

SHOW ALL WORK FOR FULL CREDIT. ***Non-Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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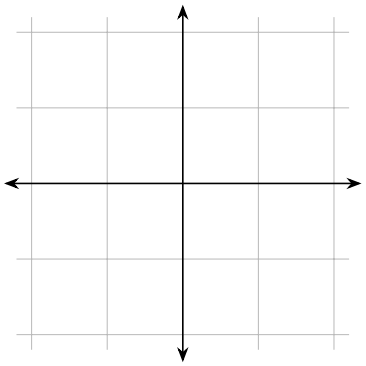
_____ _____

Signature

Write the pair of parametric equations in rectangular form, including the domain. Then sketch the curve, including direction. (3 points each)

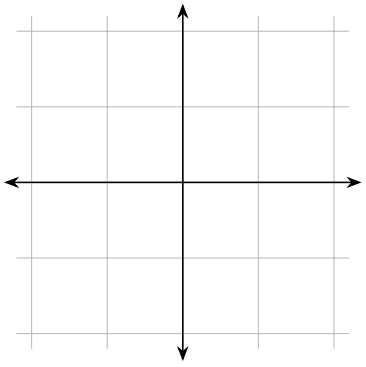
1. $x = t - 3, \quad y = -t^2 + 6t - 9, \quad 1 \le t \le 5$

SIMPLIFIED rectangular form: _____



2. $x = \sqrt{t}, \quad y = t + 1, \quad 0 \le t \le 4$

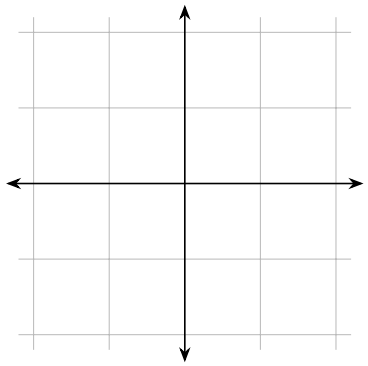
SIMPLIFIED rectangular form: _____



Write the pair of parametric equations in rectangular form. Then sketch the curve. (3 points each)

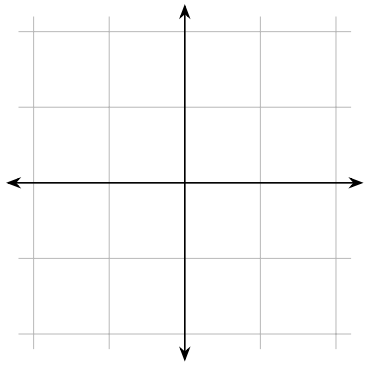
3. $x = 4\cos t, \quad y = 2\sin t, \quad 0 \le t \le 2\pi$

rectangular form: _____



4. $x = 4\tan t, \quad y = 2\sec t, \quad t \text{ in } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

rectangular form: _____



Convert the rectangular equation to parametric form. Give two different sets of answers. (2 points)

5. $y = 2(x-3)^2 - 1$

A sequence is defined recursively. Write the first five terms. (2 points)

6. $a_1 = -1; \quad a_2 = 3; \quad a_n = a_{n-1} \cdot a_{n-2}$

Extra practice: the other sequence questions #6 could be

Not part of the 16 points. One of these types is the one sequence question; worked in the key.

Write down the first five terms of the sequence.

E1. $\{a_n\} = \left\{\dfrac{n-1}{n+1}\right\}$

Write out the sum.

E2. $\displaystyle\sum_{k=1}^{n} \frac{k^2}{k+1}$

Express the sum using summation notation.

E3. $2+4+8+16+\cdots+1024$

Answer Key

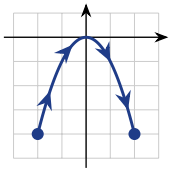
Work on the left, the reason beside it. Boxed = final answer.

Day 1: Non-Calculator

1. $x = t - 3,\ y = -t^2 + 6t - 9,\ 1 \le t \le 5$

t	(x, y)
1	$(-2, -4)$
2	$(-1, -1)$
3	$(0, 0)$
4	$(1, -1)$
5	$(2, -4)$

brackets: plug in both endpoints



left to right; solid dots, no end arrows (closed interval)

$t = x + 3$ — *isolate t in x: one answer*

$y = -(x+3)^2 + 6(x+3) - 9$ — *substitute*

$y = -x^2 - 6x - 9 + 6x + 18 - 9$ — *SIMPLIFIED: get y alone and simplify*

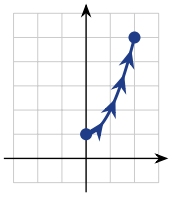
$\boxed{y = -x^2 \text{ for } x \text{ in } [-2, 2]}$ — *domain from the x column*

Other way: $y = -(t-3)^2$ and $t - 3 = x$, so $y = -x^2$.

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2. $x = \sqrt{t},\ y = t + 1,\ 0 \le t \le 4$

t	(x, y)
0	$(0, 1)$
1	$(1, 2)$
4	$(2, 5)$



starts at $(0,1)$, ends at $(2,5)$

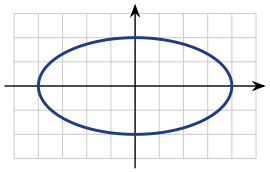
$x^2 = t$ — *square: one answer for t*

$\boxed{y = x^2 + 1 \text{ for } x \text{ in } [0, 2]}$ — *$x = \sqrt{t}$ is never negative: right half only*

3. $x = 4\cos t,\ y = 2\sin t,\ 0 \le t \le 2\pi$

t	(x,y)
0	$(4,0)$
$\frac{\pi}{2}$	$(0,2)$
π	$(-4,0)$
$\frac{3\pi}{2}$	$(0,-2)$
2π	$(4,0)$

quadrant values



direction not asked here

$\cos t = \frac{x}{4},\quad \sin t = \frac{y}{2}$ — *solve each for the trig function*

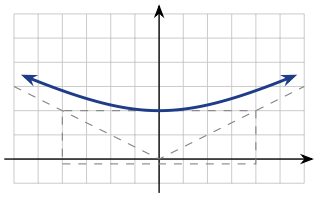
$\left(\frac{x}{4}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$ — $\sin^2 t + \cos^2 t = 1$

$\boxed{\frac{x^2}{16} + \frac{y^2}{4} = 1}$ — *whole ellipse: no domain, done*

4. $x = 4\tan t,\ y = 2\sec t,\ t$ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

t	(x,y)
$-\frac{\pi}{4}$	$(-4, 2\sqrt{2})$
0	$(0,2)$
$\frac{\pi}{4}$	$(4, 2\sqrt{2})$

parentheses: ends undefined



top branch only; end arrowheads: it keeps going

$\tan t = \frac{x}{4},\quad \sec t = \frac{y}{2}$

$\frac{y^2}{4} - \frac{x^2}{16} = 1$ — $\sec^2 t - \tan^2 t = 1$

$y^2 = 4\left(\frac{x^2}{16} + 1\right)$ — *only the top branch ($\sec t > 0$); domain can't cut it, so solve for y*

$\boxed{y = 2\sqrt{\frac{x^2}{16} + 1}}$ — *keep +; never "$y \ge 2$"*

Also fine: $y = \sqrt{\frac{x^2}{4} + 4}$. *Any equivalent form.*

5. $y = 2(x-3)^2 - 1$

$\boxed{x = t,\ y = 2(t-3)^2 - 1}$

$x = t+3:\ y = 2(t+3-3)^2 - 1$ — *pick $t+3$ so the inside is t*

$\boxed{x = t+3,\ y = 2t^2 - 1}$

Both for t in $(-\infty, \infty)$ (assumed; writing it is fine). Give both $x =$ and $y =$. Don't expand.

6.

$a_1 = -1,\ a_2 = 3$ — *both given; the formula starts at $n = 3$*

$a_3 = a_2 \cdot a_1 = 3(-1) = -3$ — *previous term times the one before it*

$a_4 = a_3 \cdot a_2 = (-3)(3) = -9$

$a_5 = a_4 \cdot a_3 = (-9)(-3) = 27$

$\boxed{-1,\ 3,\ -3,\ -9,\ 27}$ — *a list with commas, no ellipsis*

Extra practice (not scored)

E1. $\left\{\frac{n-1}{n+1}\right\}$

$\frac{0}{2}, \frac{1}{3}, \frac{2}{4}, \frac{3}{5}, \frac{4}{6}$ — *plug in $n = 1$ to 5*

$\boxed{0, \frac{1}{3}, \frac{1}{2}, \frac{3}{5}, \frac{2}{3}}$ — *$\frac{0}{2} = 0$; reduce $\frac{2}{4}$ and $\frac{4}{6}$*

E2. $\sum_{k=1}^{n} \frac{k^2}{k+1}$

$\frac{1^2}{2} + \frac{2^2}{3} + \frac{3^2}{4} + \cdots + \frac{n^2}{n+1}$ — *$k = 1, 2, 3$, then $k = n$*

$\boxed{\frac{1}{2} + \frac{4}{3} + \frac{9}{4} + \cdots + \frac{n^2}{n+1}}$ — *don't add*

E3. $2 + 4 + 8 + 16 + \cdots + 1024$

$2^1, 2^2, 2^3, 2^4, \ldots, 2^{10}$ — *powers of 2;* $1024 = 2^{10}$

$\boxed{\sum_{k=1}^{10} 2^k}$ — *lower 1, upper 10*